

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely log in using a username and password.</i>	High
2	Authorization levels	<i>Admin can manage datasets and models; users can enter data and view predictions.</i>	High

3	External interfaces	<i>Uses HDI dataset (e.g., Kaggle/UNDP) and visualization libraries such as Plotly/Matplotlib.</i>	Medium
4	Transactions processing	<i>Validates user input, preprocesses data, runs the ML model, and stores prediction results.</i>	High
5	Reporting	<i>Displays predicted HDI value, charts, and allows report download.</i>	High
6	Business rules, etc.	<i>Accept only valid indicator values; generate predictions only after successful validation.</i>	High
7	Compliance to laws or regulations	<i>Protect user data and follow basic data privacy practices.</i>	Medium
8	Other	<i>Maintain prediction history for future analysis.</i>	Low

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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Step 2: Non-Functional Requirements (NFR)

Performance & Speed	<i>Generate predictions quickly.</i>	Prediction in less than 3 seconds
Scalability	<i>Support multiple users and larger datasets.</i>	<i>Handles 100+ concurrent users)</i>
Security & Data Privacy	<i>Protect user information and stored data.</i>	<i>Password encryption and secure data storage</i>
Reliability & Availability	<i>Ensure stable operation with minimal downtime.</i>	<i>99% system availability</i>
Usability & Accessibility	<i>Provide a simple, responsive, and user-friendly interface.</i>	<i>Easy navigation on desktop and mobile</i>

<i>Other</i>	<i>Maintainability</i>	Modular code with proper documentation for easy updates
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